PennState College of Engineering
HAROLD AND INGE MARCUS DEPARTMENT OF INDUSTRIAL AND MANUFACTURING ENGINEERING

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Multi-Horizon Stochastic Programming

What is Stochastic Programming?

Stochastic programming (SP) is a mathematical modeling and optimization technique that seeks optimal decisions under *uncertainty*.

- Deterministic vs Stochastic
  - Deterministic: All data well known. (i.e. Linear Programming)
  - Stochastic: Some data are random.

The field, also known as *optimization under uncertainty*, is developing rapidly with contributions from many disciplines such as **operations research, economics, mathematics, probability, and statistics**.

General Formulation for SP

$$\begin{aligned} \min \quad & c^T x + \mathbb{E}_\xi Q(x, \xi) \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0 \end{aligned}$$

, where x is *first stage decision (strategic decision)*, ξ is random vector indicate uncertainties. $\mathbb{E}_\xi$ denote mathematical expectation with respect to ξ.

$$\begin{aligned} Q(x, \xi) = \min \quad & q^T y \\ \text{s.t.} \quad & Wy = h - Tx \\ & x, y \geq 0 \end{aligned}$$

, where y is *second stage decision (operational decision)*. ξ is the vector formed by the components of q^T , h^T , and T

Multi-stage vs. Multi-horizon

Concepts

- **Multi-stage Stochastic Programming (MSSP)**
  - *Definition:* Framework for decision-making across multiple time periods.
  - *Key Focus:* Decisions at each stage depend on **prior decisions** and **random factors**.

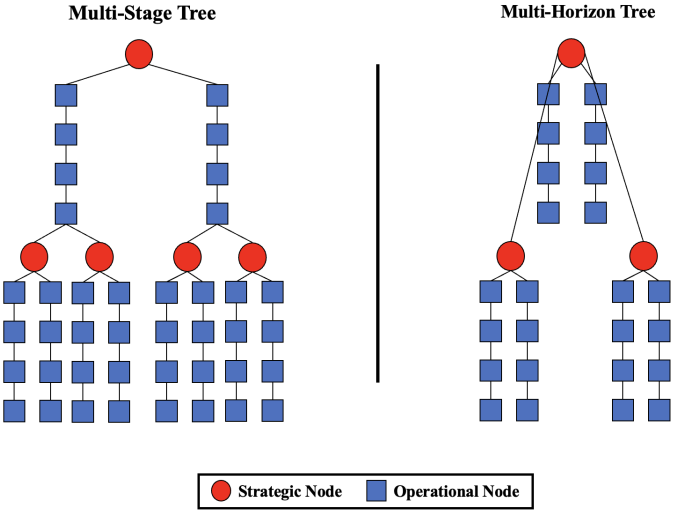
- **Multi-horizon Stochastic Programming (MHSP)**
  - *Definition:* Decision-making focused on a specific time horizon.
  - *Key Focus:* Optimizing decisions within a defined **time span**.

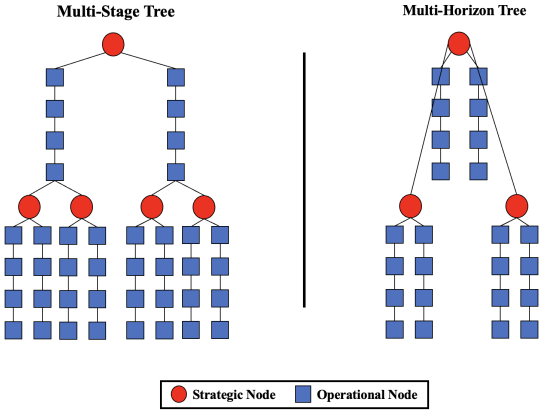
Commonalities and Differences

- **Commonalities:**
  - Both deal with decision-making under uncertainty.
  - Time plays a pivotal role in structuring decisions.

- **Differences:**
  - *Stages vs. Horizon:* Multi-stage emphasizes decisions across **continuous periods**, while Multi-horizon focuses on a **specific duration**.
  - *Decision Sequence:* Multi-stage involves interlinked decisions, where one affects the next, while Multi-horizon might be more holistic for its duration.

Structure of MHSP

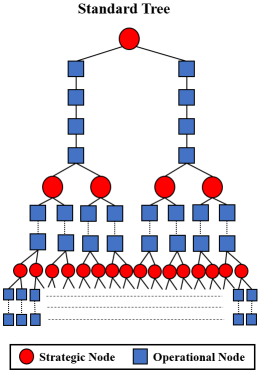


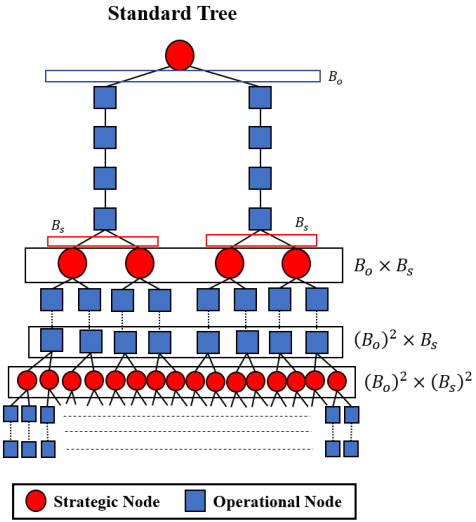


- Features
  1. MSSP and MHSP are on the same period.
  2. MSSP tree have connection between last operational node and subsequent strategic decision while MHSP have not.
  3. MSSP have more decision (node) than MHSP.

Curse of Dimensionality (Multi-Stage Tree)

- B_s, B_o denote the number of options are strategic and operational node can have.
- P_s, P_o denote the decision time period for strategic and operational node.
  - Following tree have $P_s = 3, P_o = 4, B_s = 2, B_o = 2$





- Based on the following figure, we can easily understand the pattern of the number of strategic nodes. Based on these patterns, we can define the number of strategic nodes as follows:

$$\begin{aligned} &\#\text{ of strategic nodes} \\ &= 1 + (B_o B_s)^1 + (B_o B_s)^2 \end{aligned}$$

- If, $P_s = n, n \geq 2$,

$$\begin{aligned}\#\text{ of strategic nodes} &= 1 + (B_oB_s)^1 + (B_oB_s)^2 + \cdots + (B_oB_s)^{n-1} \\ &= \sum_{a=1}^{n} (B_oB_s)^{a-1}\end{aligned}$$

- More specifically,

$$\begin{aligned}\sum_{a=1}^{P_s} (B_oB_s)^{a-1} &= \frac{(B_oB_s)^0((B_oB_s)^{P_s} - 1)}{B_oB_s - 1} \\ &= \frac{(B_oB_s)^{P_s} - 1}{B_oB_s - 1}\end{aligned}$$

- The same logic can be applied to estimate the number of operational nodes in the standard tree.
- Operational nodes are highly impacted by B_o and P_o. $B_o \times P_o$ indicates the number of operational nodes derived from one strategic node.
- We can estimate the number of operational nodes in the standard tree as follows:

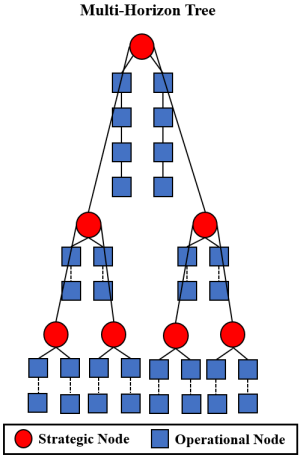
$$
\begin{aligned}
\text{\# of operational nodes} &= B_oP_o \times (\text{\# of strategic node at stage 1}) \\
&+ B_oP_o \times (\text{\# of strategic node at stage 2}) \\
&+ B_oP_o \times (\text{\# of strategic node at stage 3}) \\
&+ \cdots \\
&= \sum_{n=1}^{P_s} B_oP_o \times (\text{\# of strategic node at stage n}) \\
&= \sum_{n=1}^{P_s} B_oP_o \times (B_oB_s)^{n-1}
\end{aligned}
$$

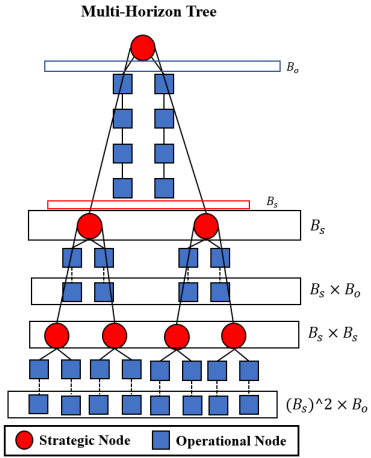
- More specifically,

$$\begin{aligned}\sum_{n=1}^{P_s} B_o P_o \times (B_o B_s)^{n-1} &= B_o P_o \sum_{n=1}^{P_s} (B_o B_s)^{n-1} \\ &= B_o P_o \times \left(\frac{(B_o B_s)^{P_s} - 1}{B_o B_s - 1}\right) \\ &= B_o P_o \frac{(B_o B_s)^{P_s} - 1}{B_o B_s - 1}\end{aligned}$$

Curse of Dimensionality (Multi-Horizon Tree)

- Let's move on to Multi-horizon tree.
- Under the same conditions as the standard tree: $P_s = 3, P_o = 4, B_s = 2, B_s = 2$.





- Unlike the standard tree, the number of strategic nodes is not affected by the number of options for the operational decision, B_o.
- Thus, the number of strategic nodes changes based on B_s.

$$\#\text{ of strategic nodes} = 1 + B_s + B_s^2$$

- If, $P_s = n, n \geq 2$,

$$\begin{aligned} \#\text{ of strategic nodes} &= 1 + (B_s)^1 + (B_s)^2 + \cdots + (B_s)^{n-1} \\ &= \sum_{a=1}^{n} (B_s)^{a-1} \end{aligned}$$

- More specifically,

$$\begin{aligned} \sum_{a=1}^{P_s} (B_s)^{a-1} &= \frac{(B_s)^0((B_s)^{P_s} - 1)}{B_s - 1} \\ &= \frac{(B_s)^{P_s} - 1}{B_s - 1} \end{aligned}$$

- Likewise, the same logic can be applied to estimate the number of operational nodes in the multi-horizon tree.
- We can estimate the number of operational nodes in the multi-horizon tree as follows:

$$
\begin{aligned}
\text{\# of operational nodes} &= B_o P_o \times (\text{\# of strategic node at stage 1}) \\
&+ B_o P_o \times (\text{\# of strategic node at stage 2}) \\
&+ B_o P_o \times (\text{\# of strategic node at stage 3}) \\
&+ \cdots \\
&= \sum_{n=1}^{P_s} B_o P_o \times (\text{\# of strategic node at stage n}) \\
&= \sum_{n=1}^{P_s} B_o P_o \times (B_s)^{n-1}
\end{aligned}
$$

- More specifically,

$$\begin{aligned}\sum_{n=1}^{P_s} B_o P_o \times (B_s)^{n-1} &= B_o P_o \sum_{n=1}^{P_s} (B_s)^{n-1} \\ &= B_o P_o \times \left(\frac{(B_s)^{P_s} - 1}{B_s - 1}\right) \\ &= B_o P_o \frac{(B_s)^{P_s} - 1}{B_s - 1}\end{aligned}$$

Table 1 Numbers of strategic and operational nodes for the standard tree in Fig. 2c and the multi-horizon tree in Fig. 3b

	Standard tree	Multi-horizon tree
# strategic nodes	$\frac{1-(B_o B_s)^{P_s}}{1-B_o B_s} \approx (B_o B_s)^{P_s}$	$\frac{1-(B_s)^{P_s}}{1-B_s} \approx B_s^{P_s}$
# operational nodes	$P_o B_o \frac{1-(B_o B_s)^{P_s}}{1-B_o B_s} \approx P_o B_o (B_o B_s)^{P_s}$	$P_o B_o \frac{1-(B_s)^{P_s}}{1-B_s} \approx P_o B_o B_s^{P_s}$

P_s is the number of strategic periods, P_o the number of operational periods per strategic period. B_s and B_o are the numbers of strategic and operational successors in each strategic and operational branching, respectively. We assume that these numbers are constant throughout the tree and that $B_s > 1$

- The number of decisions (nodes) for multi-horizon tree are less than multi-stage tree in any case.
- The number of decisions (nodes) are exponentially increased in both trees.
- We can avoid curse of dimensionality by using multi-horizon approach, because we have fewer decision than multi-stage approach during same period.

EMPIRE model

What is EMPIRE model?

Background

- **Goal**: Reduce EU greenhouse gas emissions by 80–95% from 1990 levels by 2050.
  - Almost complete decarbonization of the power sector required.
  - Emphasis on large-scale deployment of wind and solar power.
- **Challenges**: Intermittency; something that starts, then stops, then starts up again and balancing supply-demand due to wind and solar.
- **Importance**: Consider short-term uncertainties in planning investments in generation, transmission, and storage.

EMPIRE model

- **EMPIRE** (**E**uropean **M**odel for **P**ower system **I**nvestment with **R**enewable **E**nergy) is a *stochastic programming* model, developed to handle the challenges related to intermittent energy production and stochastic energy demand in a long-term investment model.

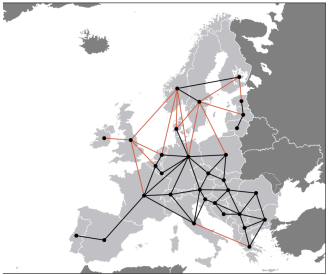


Figure: Spatial detail of the EMPIRE model

- A total of 31 European countries, connected through 55 interconnectors.
- Capacity expansion model designed to assess optimal capacity investments and system operation over **40–50 years.**
- Central planner perspective (i.e. government control, not companies)

Objective

- **Minimize** the sum of *investment* and *(expected) operational* costs for whole of the system, over all time periods.

Approach

- *Multi-Horizon Stochastic Programming*; Kaut et al. (2014)
  - Why? : To avoid the *curse of dimensionality* when modeling short-term uncertainty in a long-term model
  - How? : Simultaneously handles short-term dynamics, short-term uncertainty as well as long-term dynamics.

Dynamics

1. long-term dynamics: multiple investment periods
2. short-term dynamics: multiple sequential operational decision periods
3. short-term uncertainty: multiple scenarios for input data describing operating conditions
   (wind, solar and load profiles, hydro power production limits, etc.)

Uncertainty

- **Strategic uncertainty**: Assumed perfect foresight in terms of the strategic data. (No uncertainty in strategic decision)
- **Operational uncertainty**:
  - load profiles
  - wind and solar generation profiles
  - seasonal availability of water stored in reservoir for hydroelectric production.

Mathematical Formulation

Objective Function

- Minimize the sum of investment(x) and expected operational(y_ω) costs for the system

$$\min_{\mathbf{x},\{y_\omega\}_{\omega\in\Omega}} z = \sum_{i\in\mathcal{I}}(1+r)^{-5(i-1)}\times$$
$$\left\{\sum_{g\in\mathcal{G}} c_{gi}^{\text{gen}} x_{gi}^{\text{gen}} + \sum_{l\in\mathcal{L}} c_{li}^{\text{tran}} x_{li}^{\text{tran}} + \sum_{b\in\mathcal{B}}\left(c_{bi}^{\text{storPW}} x_{bi}^{\text{storPW}} + c_{bi}^{\text{storEN}} x_{bi}^{\text{storEN}}\right)\right.$$
$$\left.+\mathcal{V}\sum_{\omega\in\Omega}\pi_\omega\sum_{s\in\mathcal{S}}\alpha_s\sum_{h\in\mathcal{H}}\sum_{n\in\mathcal{N}}\left[\sum_{g\in\mathcal{G}_n} q_{gi}^{\text{gen}} y_{ghi\omega}^{\text{gen}} + q_{ni}^{\text{ll}} y_{nhi\omega}^{\text{ll}}\right]\right\}$$

, where $\mathcal{I}$ denotes time periods, r denotes discounted rate, $\mathcal{G}, \mathcal{L}$ and $\mathcal{B}$ denote set of generators, interconnectors and storages respectively.

Constraints for operational variables

1. Energy Balance Constraints
2. Capacity balance constraints
3. Maximum production constraints
4. Production for thermal generators constraints
5. Storage energy balance constraints
6. Storage energy limit constraints
7. Hydroelectric power generation
8. Exchange capacity limitation

1. Net Energy Balance

- The sum of net local generation and net import is required to balance the **net load** at every node (n), hour (h), period (i).

$$\sum_{g\in\mathcal{G}} y^{\text{gen}}_{ghi\omega} + \sum_{b\in\mathcal{B}_n} \eta^{\text{discharg}}_{b} y^{\text{discharg}}_{bhiw} - y^{\text{charg}}_{bhi\omega} + \sum_{a\in\mathcal{A}^{in}_n} \eta^{\text{tran}}_{a} y^{\text{flow}}_{ahi\omega} - \sum_{a\in\mathcal{A}^{out}_n} y^{\text{flow}}_{ahi\omega} = \xi^{\text{load}}_{\mathbf{nhi}\omega} - \mathbf{y}^{\mathbf{ll}}_{\mathbf{nhi}\omega}$$

, where $n \in \mathcal{N}$, $h \in \mathcal{H}$, $i \in \mathcal{I}$ and $\omega \in \Omega$

- η: efficiency parameter
  - $\eta^{\text{discharg}}_{b}, \eta^{\text{tran}}_{a} \in (0, 1)$

2. Installments Capacity balance

- Capacity balance for generator, interconnector, storage (power and energy).
- For instance, the capacity of generator at the beginning of period i;

$$v_{gi}^{gen} = \bar{x}_{gi}^{gen} + \sum_{j=i'}^{i} x_{gj}^{gen}, \quad i' = \max\{1, i - \lfloor i_g^{\text{life}}/5 \rfloor\}, i \in \mathcal{I}, g \in \mathcal{G}$$

- v_{gi}^{gen}: maximum installed capacities for generator ($g \in \mathcal{G}$) at the beginning of period i
- $\bar{x}_{gi}^{gen}$: existing capacity at the beginning of period i

We can balance for interconnector,storage (power and energy) in the same way.

How can we interpret those balance equation?

- $i_g^{\text{gen}} = 18$ years
- $\mathcal{I} = \{1, 2, 3, \ldots, 8\}$, $i \in \mathcal{I}$ denote five year time block.

Then, $i^{'} = max\{1, i - \lfloor 18/5 \rfloor\} = max\{1, i - 3\}$, So until $i = 4$, $i^{'}$ is to be 1.

$$v_{g1}^{gen} = \bar{x}_{g1}^{gen} + \sum_{j=1}^{1} x_{gj}^{gen}$$

$$v_{g2}^{gen} = \bar{x}_{g2}^{gen} + \sum_{j=1}^{2} x_{gj}^{gen}$$

$$\vdots$$

$$v_{g4}^{gen} = \bar{x}_{g4}^{gen} + \sum_{j=1}^{4} x_{gj}^{gen}$$

$$v_{g5}^{gen} = \bar{x}_{g5}^{gen} + \sum_{j=2}^{5} x_{gj}^{gen}$$

More specifically,

$$
\begin{aligned}
(i=1)\ \ & v_{g1}^{gen} = \bar{x}_{g1}^{gen} + (x_{g1}^{gen}) \\
(i=2)\ \ & v_{g2}^{gen} = \bar{x}_{g2}^{gen} + (x_{g1}^{gen} + x_{g2}^{gen}) \\
& \vdots \\
(i=4)\ \ & v_{g4}^{gen} = \bar{x}_{g4}^{gen} + (x_{g1}^{gen} + x_{g2}^{gen} + x_{g3}^{gen} + x_{g4}^{gen}) \\
(i=5)\ \ & v_{g5}^{gen} = \bar{x}_{g5}^{gen} + (x_{g2}^{gen} + x_{g3}^{gen} + x_{g4}^{gen} + x_{g5}^{gen})
\end{aligned}
$$

When $i = 5$ (e.g. 21-25 years), x_{g1}^{gen} is disappeared. Because x_{g1}^{gen} is over $i_g^{\mathrm{gen}} = 18$ when it comes to $i = 5$.

3. Maximum production constraints

- Intermittent production, such as wind power and solar power, the availability parameters are stochastic.
- Production from each generator ($\mathcal{G}$) for a hour ($\mathcal{H}$) is limited by the available installed capacity.

$$y^{\text{gen}}_{ghi\omega} \leq \xi^{\text{gen}}_{ghi\omega} v^{\text{gen}}_{gi}$$

, where $g \in \mathcal{G}, h \in \mathcal{H}, i \in \mathcal{I}, \omega \in \Omega$

- $\xi^{\text{gen}}_{ghi\omega}$: Availability parameters
  - $\xi^{\text{gen}}_{ghi\omega} \in (0, 1)$

4.Ramp up of production for thermal generators

- limited to a certain share of the installed capacity

$$|y^{\text{gen}}_{ghi\omega} - y^{\text{gen}}_{g(h-1)i\omega}| \leq \gamma^{\text{gen}}_{g} v^{\text{gen}}_{gi}$$

, where $g \in \mathcal{G}^{\text{Thermal}}, s \in \mathcal{S}, h \in \mathcal{H}^{-}_{s}, i \in \mathcal{I}, \omega \in \Omega$

$$-\gamma^{\text{gen}}_{g} v^{\text{gen}}_{gi} \leq y^{\text{gen}}_{ghi\omega} - y^{\text{gen}}_{g(h-1)i\omega} \leq \gamma^{\text{gen}}_{g} v^{\text{gen}}_{gi}$$

5. Storage energy balance constraints

- At every hour (h), the storage end-level (w) is set as the difference between the energy level in the previous hour ($h-1$) minus the net discharge.

$$w^{\text{stor}}_{b(h-1)i\omega} + \eta^{\text{chrg}}_{b} y^{\text{chrg}}_{bhi\omega} - y^{\text{dischrg}}_{bhi\omega} = w^{\text{stor}}_{bhi\omega}$$

, where $b \in \mathcal{B}, s \in \mathcal{S}, h \in \mathcal{H}^{-}_{s}, i \in \mathcal{I}, \omega \in \Omega$

- $w^{\text{stor}}_{bhi\omega}$: decision variables keep track of the energy level for storage $b \in \mathcal{B}$

6. Energy limit

- The stored energy and charging/discharging are limited by the energy and power installed capacities.

$$
\begin{aligned}
w^{\text{stor}}_{b(h-1)i\omega} &\leq v^{\text{stor}}_{bi} \\
y^{\text{chrg}}_{bhi\omega} &\leq v^{\text{storPW}}_{bi} \\
y^{\text{dischrg}}_{bhi\omega} &\leq \rho_b v^{\text{storPW}}_{bi}
\end{aligned}
$$

, where $b \in \mathcal{B}, h \in \mathcal{H}, i \in \mathcal{I}, \omega \in \Omega$

- ρ_b: non-negative parameter, determines a fixed discharge to charge power capacity ratio.

In simple terms, ρ_b represents the relationship between how fast a battery can be charged and how fast it can discharge its energy.

For example:

- $\rho_b = 1$
  - It means that the maximum charging power and the maximum discharging power of the battery are the **same**.
  - In other words, the time it takes to fully charge the battery would be the same as the time it takes to fully discharge it.
- $\rho_b = 0.5$
  - It means that the maximum charging power is **only half** of the maximum discharging power.
  - In this case, it would take twice as long to fully charge the battery compared to fully discharging it.

7. Hydroelectric power generation

- Water availability in reservoirs and hydroelectric resource potential.
- For regulated hydroelectric generators, energy limits constrain the total production over each season.

$$\sum_{h \in \mathcal{H}_s} y^{\text{gen}}_{ghi\omega} \leq \xi^{\text{RegHydroLim}}_{gsi\omega}$$

- For every node, both regulated and unregulated is limited in terms of annual energy production.

$$\sum_{s \in \mathcal{S}} \alpha_s \times \sum_{h \in \mathcal{H}_s} \sum_{g \in \mathcal{G}^{\text{Hydro}}_n} y^{\text{gen}}_{ghi\omega} \leq \xi^{\text{HydroLim}}_{ni\omega}$$

8. Exchange capacity limitation

- EMPIRE has a simplified network description, only considering import/export links between countries.
- Exchange is limited by the (symmetric) capacity for each interconnector.

$$y^{\text{flow}}_{ahi\omega} \leq v^{\text{tran}}_{li}$$

Constraints for strategic variables

1. limits on possible investments per period for a given technology in a given node

$$\sum_{g \in \mathcal{G}_{tn}} x_{gi}^{\text{gen}} \leq \bar{X}_{ti}^{\text{gen}}$$

, where $t \in \mathcal{T}^{\text{AggTech}}, n \in \mathcal{N}, i \in \mathcal{I}$

2. maximum installed capacity constraints

$$\sum_{g \in \mathcal{G}_{tn}} v_{gi}^{\text{gen}} \leq \bar{V}_{ti}^{\text{gen}}$$

, where $t \in \mathcal{T}^{\text{AggTech}}, n \in \mathcal{N}, i \in \mathcal{I}$

- lines and storages (power and energy) capacity investments and installed capacities

$$v_{bi}^{\mathrm{storPW}} - \beta_b v_{bi}^{\mathrm{storEN}} = 0$$

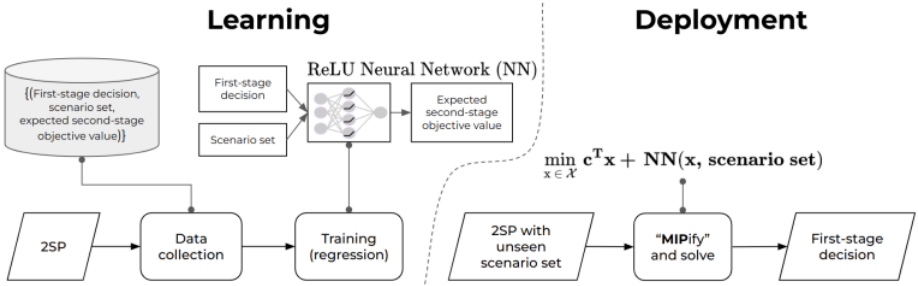
, where $b \in \mathcal{B}^+, i \in \mathcal{I}$

- $\mathcal{B}^+$: subset of storages, such as batteries
- β_b: fixed storage power to energy ratio.

Future Work

Future work

1. Build a neural network to approximate the expected second stage value. (Neur2SP)



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